Kaushalendra Singh

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SUMMARY

AI and backend engineer who builds **LLM agents, RAG systems, MCP servers and evaluation pipelines** at an AI startup, alongside production **Python (FastAPI)** and **Node.js** backends. Cut P95 API latency by 50%, shipped products serving 1300+ users, launched an AI developer tool on Product Hunt, placed in hackathons (winner and runner-up), and have merged pull requests in a 54K-star open-source repository.

EXPERIENCE

EpicWise
Remote (Canada Startup)

AI Founding Engineer Intern — *May 2026 – Present*

- **Helped the founding team ship EpicWise's core context engine faster** by designing multi-step retrieval and reasoning pipelines using LangGraph.
- **Enabled secure third-party integrations with EpicWise** by building its public MCP server, including authentication, authorization, an agent harness testing agent, and a requirement-analysis agent.
- **Improved answer accuracy for complex user queries** by implementing a hybrid retrieval pipeline combining vector search (Pinecone) and graph-based retrieval (Neo4j), and authoring the system prompts, execution routers, and tool-calling schemas behind it.
- **Reduced production LLM errors** by building an evaluation pipeline that tracked tool correctness, answer relevancy, correctness, and hallucination metrics across agent runs using DeepEval, catching regressions before deployment.

GRS Worker
Remote

Full Stack Developer Intern — *Sep 2025 – Nov 2025*

- **Supported reliable performance under concurrent load** by building a scalable backend using Node.js, Prisma, and PostgreSQL with Redis caching and secure authentication.
- **Reduced data errors by 40%** by implementing Zod validation and structured service-layer workflows.
- **Streamlined service deployment and customer communication** by integrating Twilio and QR-based systems, then containerizing and deploying all services with Docker and CI/CD pipelines.
- **Cut P95 API latency from ∼800ms to ∼400ms (50% reduction)** by optimizing database queries and the Redis caching strategy.

ACHIEVEMENTS

- **2nd place, Tambo AI Hackathon (UI Strikes Back):** designed and built an AI-driven system with a focus on usability and system design, under time constraints.
- **Hackathon winner, CosmoDetect:** a YOLOv11 and PyTorch object-detection API served with FastAPI and an OpenCV preprocessing pipeline, deployed as a live API.
- **Open source, Joplin (54K+ GitHub stars):** 2 merged PRs (a renderer crash on closing a secondary window, and OneNote zip import) and an open PR adding OAuth for Joplin Server; built NoteGraph, an AI knowledge-graph plugin shared with the Joplin community (proof of work).
- **Open source, REA and Mesher organizations:** contributor, collaborating with maintainers on their repositories.

PROJECTS

Buildify Labs: Agentic AI Business Intelligence
Live | *GitHub*

React, Vite, FastAPI, Groq, HuggingFace, PostgreSQL, pgvector, Zustand

- **Turned plain-English questions into SQL-driven charts and explanations** by building an AI BI copilot on a judge-narrate-guarantee pipeline with LLM tool routing, sufficiency judging, and deterministic visual synthesis with grounding checks.
- **Grounded answers in each user's own documents** by implementing RAG over uploaded PDFs: text extraction and chunking, HuggingFace embeddings, and per-user-scoped pgvector cosine retrieval, returned as validated JSON and streamed live over SSE.
- **Made multi-turn conversations resolve without re-asking** by adding thread-scoped prior-turn context and persisted chat history, with context-window budgeting through row truncation, evidence ranking, and long-source summarization.

RepoGraph AI: VS Code Extension (57+ Downloads, 4000+ Visitors)
Live | *GitHub*

TypeScript, VS Code API, Groq, Ollama, Gemini, Anthropic, OpenAI

- **Helped developers understand unfamiliar codebases faster** by building a VS Code extension that scans workspaces and maps every file relationship into an interactive, force-directed dependency graph.
- **Gave users flexibility to choose their own AI provider** by integrating 5 AI providers (Groq, Ollama, Gemini, Anthropic, OpenAI) with custom model support and zero telemetry.
- **Made codebase Q&A more reliable** with a multi-turn agent using full graph, summary, and file context, plus smart filtering across 20+ languages.
- **Launched on Product Hunt and listed on the XERJ Corpus Hub** as a contributed repository, with a public contributor page.

Evident: Open-Source Extraction Layer for AI Agents
Live | *GitHub*

Python, MCP, REST API, Crawl4AI, Playwright, pytest, GitHub Actions

- **Gave AI agents a way to judge how much to trust extracted data** by building an open-source extraction and fetch layer that attaches a confidence score and method to every result, callable via MCP, REST, or a native Python SDK.
- **Made reliable extraction possible on sites without a hand-written parser** by implementing a tiered resilience ladder, static fetch escalating to JS rendering via Crawl4AI/Playwright, plus a versioned, community-contributable recipe system for sites like Greenhouse, Lever and Ashby.
- **Protected the fetch layer from server-side request forgery** by building an SSRF guard that resolves hostnames to IPs and blocks private, loopback, and cloud-metadata address ranges before any connection is opened.

MyAttendance
Live | *GitHub*

Next.js, MongoDB, Redis, PWA

- **Served 1300+ users reliably** by building a PWA with role-based access control and MongoDB queries optimized for high concurrency.
- **Reduced database load and improved uptime visibility** with a Redis caching layer and structured error logging for production monitoring.

EDUCATION

Harcourt Butler Technical University, Kanpur
Bachelor of Technology — *CGPA: 7.0 / 10*

TECHNICAL SKILLS

- **Languages:** JavaScript/TypeScript, Python, SQL, C++
- **Agentic Systems:** LangChain, LangGraph; ReAct & Plan-Execute patterns; tool calling; multi-step reasoning workflows; MCP servers and auth
- **RAG & Retrieval:** Embeddings, Pinecone/FAISS/pgvector vector DBs, Neo4j graph retrieval, semantic search, context & memory management
- **LLM Systems:** OpenAI/Gemini/Anthropic APIs, structured outputs, prompt engineering, state handling in AI workflows; PyTorch fine-tuning
- **Evaluation:** DeepEval for LLM output evaluation, regression testing, quality benchmarking and agent harness testing of AI agent responses
- **Frontend:** Next.js, React, Tailwind, Zustand
- **Backend:** Node.js, Express, FastAPI, Prisma, REST/GraphQL, Kafka, Redis
- **Database:** MongoDB, PostgreSQL, MySQL
- **DevOps:** Docker, Kubernetes, AWS, CI/CD, Logging & Monitoring